SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	ERROR 303
Project / Solution Name	StockWise
Name (Leader)	Timothy Lai Yong Fong
Email & Phone (Leader)	timothyfy851@gmail.com / +601113258211
University Affiliation(s)	Taylor's University
Live Prototype URL	https://error303v2.vercel.app/
Code Repository URL (public / access granted to judges)	https://github.com/timlai0402/ERROR303-2.0.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

StockWise is a responsive inventory web application for SMEs. The React interface supports products, stock movements, FIFO batch picking, barcode/QR scanning, history, statistics and low-stock alerts. Browser-side JavaScript calculates days of stock remaining from the previous 14 days of stock-out records and supplier lead times. Reorder messages use editable templates and open WhatsApp or email links for the user to send. For Snap Delivery Note, the browser sends a compressed document image and product IDs, names and codes to the Express endpoint /api/extract-delivery-note. The server calls a configured vision model through an OpenAI-compatible or Anthropic API, keeping its API key server-side. Users review extracted quantities and product matches before saving stock-in batches. Sample notes fall back to bundled extraction results when live AI is unavailable. When configured, the browser connects directly to Supabase PostgreSQL and Realtime for persistence and synchronization; otherwise, data stays in browser storage. Vite builds the frontend. Express can serve both the built frontend and API. GitHub Pages hosts only static assets and requires a separate backend for live AI.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Node.js 18+ (required)	JavaScript server runtime	MIT
React / React DOM 18.3.1; React Router DOM 6.30.6	User interface, rendering and routing	MIT
HTML5, CSS, JavaScript (ES modules); SQL	Web structure, styling, application logic and database schema	Language standards; N/A
Supabase-js 2.118.0	Database client; server package declared as a dependency	MIT
Express 4.22.3; dotenv 16.6.1	API server and environment configuration	MIT; BSD-2-Clause

Name & Version	Purpose in Project	License
Vite 5.4.21; @vitejs/plugin-react 4.7.0	Frontend development and build	MIT
html5-qrcode 2.3.8	Camera-based QR and barcode scanning	Apache-2.0
Recharts 2.15.4; lucide-react 0.460.0	Charts and interface icons	MIT; ISC
concurrently 9.2.5	Runs frontend and backend during development	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
ChatGPT via KrackkedDev	Do picture scanning AI input feature	Sponsor-provided
Supabase API	Read/write inventory records and synchronize devices when configured	Free tier
WhatsApp wa.me and mailto links	Open user-reviewed reorder messages; no messaging API integration	Free tier

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Supabase	Persistent PostgreSQL database, authentication, and secure data storage	Free Tier, global
Github Pages	Production static web application edge hosting and continuous integration	Free Tier, global
Node.js / Express backend	Required for live document extraction; local server supported, production host to be confirmed	Free Tier, global

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
ChatGPT	Access provided by KrackkedDev	Pre-trained API inference for delivery-note extraction and product matching; no training or fine-tuning implemented	KrackkedDev access terms and applicable upstream provider terms; [confirm]

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
StockWise synthetic inventory seed	src/data/seed.js	Synthetic demo users, suppliers, products, batches and movements	Repository Apache-2.0; confirm asset provenance	N (synthetic)
Three sample delivery notes and extraction results	public/samples/; src/data/sampleNotes.js	Bundled demonstration images and fixture results	NA	N
User-entered inventory and uploaded notes	Entered or uploaded through StockWise	Private operational data; inference only	User-owned or used with permission	Y (if documents/contact fields contain personal data)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

- ☑ Yes
- ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
ChatGPT GPT-6	Provide recommendation to elevate our project idea	Moderate
Google Antigravity	Refine our website's structure and its code	Major
Claude Code	Code debugging and application structure development	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

This declaration has been checked against the project source and dependency lockfile. Before signing, the team should confirm its review of AI-generated code, FIFO and stock totals, document extraction and product matches, synchronization, scanning and reorder links. Extracted delivery-note rows are editable and require user confirmation before stock is saved.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
KrackkedDev	ChatGPT API access ($100 of credit)	Provides the AI service for delivery-note extraction and product matching

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

KrackkedDev provides access to ChatGPT for live delivery-note extraction and product matching. This API is replaceable through configurable API base URL, model and request format. Live reading of arbitrary delivery-note images requires a compatible vision service. Core inventory, FIFO, forecasting, scanning and template-based reorder drafting do not depend on a generative model. Bundled sample notes can demonstrate the review flow using pre-extracted results.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

StockWise stores business details, staff names/roles, supplier contact details, products, stock batches and timestamped movements. Data is retained in browser storage or, when configured, Supabase until reset or deleted; no scheduled retention policy is implemented. Live document extraction sends the uploaded image and product IDs, names and codes through the Express server to the configured AI provider. Images may contain personal or commercial information. The inspected server does not persist uploaded images, but provider retention depends on its terms. AI credentials are read from server environment variables. The prototype uses demo user selection and open database policies; production authentication and per-user access controls are not implemented. WhatsApp/email links pass the reviewed reorder message and recipient details to the chosen service. Use synthetic demonstration data until production access controls and retention policies are established.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Vision extraction can misread quantities, units or product names, and product matching can suggest the wrong item. StockWise provides editable review rows and requires confirmation before saving batches; low-confidence matches are flagged or treated as new items. API availability, rate limits, image quality and provider configuration affect live scanning. Bundled samples use fixed extraction results when AI is unavailable and do not prove live accuracy. Forecasts use a 14-day average rather than a trained predictive model, so sparse records or seasonal demand can distort estimates. Reorder drafts are templates, not generated model output. Demo login and permissive database policies are unsuitable for sensitive production data. No measured model-accuracy benchmark is included in the repository.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☐ Any use of official SDC sponsor resources has been declared in Section 5.
- ☐ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



1 Nov 2026